Query: What is the current stock of SKU-1042 in the Chennai warehouse, and does it match the Q3 inventory report?

IE-KRT

Evaluator

Complexity $E = 0.72$

Sensitivity $S_i = 0.76$

LLM Self-Assessment

Confidence $m_i = 0.18$

Activation Score

$K(t_i) = (E - m_i) \cdot S_i = 0.41$

$K < \theta$

No Need Retrieval

Parametric Response

Direct LLM answer, 0 ms retrieval

(35.0% of queries bypassed)

IE-KQF

$K \geq \theta = 0.25$

Need Retrieval

Internal Phrases (LLM)

SKU-1042, Chennai warehouse,

Q3 inventory report

Intent Tokens

live stock lookup +

cross-source verification

concat

Reformulated Query q^+

current stock SKU-1042 Chennai

warehouse | compare Q3 inventory report

Retrieval

Iteration

AMORA Routing

Two-Stage FAISS Funnel

$N = 10{,}000 \rightarrow M = 55$ candidates

3.26 ms | SRR = 99.45%

6-D Multi-Objective Scoring

R, L, F, C, H, K

$\Sigma_j W_j = 1$ (Pareto-optimal)

Top-K Selected Tools

inventory_db (SQL)

stock_api (REST), q3_report (DOC)

Evidence Fusion

Tool Outputs

SQL: 150 units (2 h ago)

REST: 150 units (live)

DOC: 90 units (Q3 snapshot)

Fusion Engine

Hash-window dedup: 3 → 2 facts

$O(E^2)$ conflict graph: 150 vs 90

Prompt tokens −26.7%

Fused Evidence Context

[FACT] stock = 150 (fresh, reliable)

[CONFLICT] Q3 report = 90 (stale)

provenance tags attached

Local LLM

Qwen 2.5 7B (Q4_K_M)

zero data egress

SKU-1042 currently has 150 units in the Chennai warehouse [inventory_db,

updated 2 h ago]. Note: the Q3 report lists 90 units; the gap reflects

post-Q3 restocking, so the live ledger value is authoritative.

LLM Continual Generation based on Fused Evidence